BUSINESS REQUIREMENTS DOCUMENT

Banking Business Rules Engine (BRE)

Universal BRE Platform for MFI | Retail Banking | Corporate & SME Lending

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1. Executive Summary

The Banking Business Rules Engine (BRE) is a centralized, configurable, and highly scalable platform designed to evaluate financial and operational rules across all lending verticals — Microfinance Institutions (MFI), Retail Banking, and Corporate/SME Lending. The BRE decouples business logic from application code, enabling non-technical business analysts to configure, deploy, and version-control rules without software releases.

The system supports real-time rule evaluation during loan origination, eligibility checks, deviation routing, credit scoring, fraud detection, and compliance validation. It is designed to be the single source of truth for all business decision logic across the organization.

2. Business Context & Problem Statement

2.1 Current State Problems

- Business rules are hardcoded in application services, requiring developer intervention for every change
- No versioning or audit trail for rule changes — compliance risk
- Rules are duplicated across multiple microservices leading to inconsistency
- No ability to A/B test or shadow-run new rules before production deployment
- Rule changes require full regression testing and release cycles (2-4 weeks minimum)
- No centralized dashboard to monitor rule execution outcomes across products

2.2 Desired State

- Business analysts can create, modify, and deploy rules via a web UI — no code change required
- All rule executions are logged with full audit trail for regulatory compliance
- Rules are versioned, tested in UAT, and promoted to production with one-click approval
- Real-time rule evaluation via REST API and async via Kafka events
- Rules can be simulated/back-tested against historical loan data

3. Scope of the BRE System

3.1 In Scope

Domain	Rule Categories	Examples
Microfinance (MFI)	Group eligibility, VRM category, SHG cycle checks	SHG0003, SHG0007, FOIR, NTB/ETB classification
Retail Banking	Credit score, income verification, product eligibility	CIBIL threshold, LTV ratio, EMI/NMI ratio
Corporate / SME	Financial ratio analysis, exposure limits, sector risk	DSCR, current ratio, promoter background check
Cross-cutting	Fraud, AML, KYC compliance, regulatory limits	Blacklist check, PEP screening, RBI exposure norms

3.2 Out of Scope

- Core Banking System (CBS) integration — interface only, no CBS modification
- Customer-facing UI — BRE is a backend service consumed by LOS/LMS
- Pricing engine — separate product, may consume BRE outputs

4. Stakeholders

Stakeholder	Role	Responsibility
Credit Policy Team	Rule Owner	Define and approve business rules
Technology / Engineering	Builder	Design, build, and deploy the BRE platform
Risk & Compliance	Reviewer	Validate regulatory compliance of rules
Operations / Credit Officers	Consumer	Act on BRE decisions (approve/deviate/reject)
Audit Team	Oversight	Review rule execution logs and version history

5. Functional Requirements

5.1 Rule Management

- FR-001: The system shall allow creation of rules with conditions (IF-THEN-ELSE) via a web-based rule designer
- FR-002: Rules shall support simple conditions, compound conditions (AND/OR), nested conditions, and computed expressions
- FR-003: Each rule shall have a unique rule_code, description, effective_from, effective_to, and version
- FR-004: Rules shall be categorized by product_type (MFI / RETAIL / SME), rule_category, and applicability_stage
- FR-005: Rule versions shall be immutable once activated — changes create a new version
- FR-006: Rules shall support PASS, FAIL, and DEVIATION outcomes with configurable deviation routing levels

5.2 Rule Execution Engine

- FR-007: The engine shall evaluate all applicable rules for a given loan application context in a single request
- FR-008: Rule evaluation shall complete within 500ms for up to 100 rules synchronously
- FR-009: The engine shall support both synchronous REST API and asynchronous Kafka-based execution
- FR-010: Rules shall be evaluated in a configurable execution order with dependency support

- FR-011: The engine shall support short-circuit evaluation — stop on first FAIL if configured
- FR-012: Rule inputs shall be defined as a schema; missing mandatory inputs shall return a validation error

5.3 Deviation & Escalation

- FR-013: Deviation outcomes shall specify the escalation role (CU_L3, CU_L4, CREDIT_HEAD, etc.)
- FR-014: Multiple deviations in a single application shall be aggregated and routed to the highest authority
- FR-015: Deviation expiry shall be configurable per rule — lapsed deviations auto-expire

5.4 Audit & Traceability

- FR-016: Every rule execution shall be persisted with input snapshot, output, timestamp, and triggered_by
- FR-017: Rule version history shall be fully auditable with who changed what and when
- FR-018: Execution reports shall be exportable as CSV / Excel

5.5 Rule Simulation

- FR-019: Business analysts shall be able to simulate a rule against test input data before activation
- FR-020: Back-testing shall allow running a new rule version against historical loan data in read-only mode

6. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-001	Performance	Synchronous rule evaluation < 500ms for 100 rules at P99
NFR-002	Throughput	Handle 1,000 concurrent rule evaluation requests per second
NFR-003	Availability	99.95% uptime SLA with zero-downtime deployments
NFR-004	Scalability	Horizontal scale via stateless microservice pods
NFR-005	Security	Role-based access control (RBAC) on rule create/modify/activate
NFR-006	Auditability	Full immutable audit log for every rule change and execution
NFR-007	Maintainability	New rule category onboarding without code deployment

7. Rule Types Across Banking Domains

7.1 MFI Rules

Rule Code	Rule Name	Outcome Type	Input Parameters
SHG0001	Group Formation Eligibility	PASS / FAIL	group_size, group_age_months
SHG0003	VRM Category Check	PASS / DEV / FAIL	vrm_category, loan_product
SHG0007	Cycle Type Validation	PASS / FAIL	group_cycle_type, member_count
MFI0001	FOIR (Fixed Obligation to Income)	PASS / DEV / FAIL	monthly_income, total_emi
MFI0002	Active Loan Count Check	PASS / FAIL	active_loan_count, bureau_data

7.2 Retail Banking Rules

Rule Code	Rule Name	Outcome Type	Input Parameters
RET0001	CIBIL Score Threshold	PASS / DEV / FAIL	cibil_score, product_type
RET0002	LTV Ratio Check (Secured)	PASS / DEV / FAIL	loan_amount, collateral_value
RET0003	EMI to NMI Ratio	PASS / DEV / FAIL	total_emi, net_monthly_income
RET0004	Age Eligibility	PASS / FAIL	applicant_age, loan_tenure
RET0005	Blacklist / Debarment Check	PASS / FAIL	customer_id, pan_number

7.3 Corporate / SME Rules

Rule Code	Rule Name	Outcome Type	Input Parameters
SME0001	DSCR (Debt Service Coverage)	PASS / DEV / FAIL	net_operating_income, total_debt_service
SME0002	Current Ratio	PASS / FAIL	current_assets, current_liabilities
SME0003	Promoter Background Check	PASS / FAIL	promoter_id, defaulter_list
SME0004	Sector Exposure Limit	PASS / DEV / FAIL	sector_code, current_exposure

AML0001	PEP / Sanctions Screening	PASS / FAIL / REVIEW	customer_name, pan, entity_id

8. Rule Lifecycle

Stage	Status	Description
1	DRAFT	Rule created by analyst, not yet evaluated
2	IN_REVIEW	Submitted for credit policy and risk review
3	APPROVED	Approved by rule owner, ready for UAT deployment
4	ACTIVE	Live in production, actively evaluated on requests
5	SUPERSEDED	Replaced by newer version, retained for audit history
6	DEACTIVATED	Manually disabled, no longer evaluated

9. Integration Requirements

9.1 Upstream Systems (Rule Inputs)

- Loan Origination System (LOS) — loan application data, borrower profile, group details
- Credit Bureau APIs (CIBIL, Equifax, CRIF) — credit score, active loans, delinquency history
- KYC Service — Aadhaar, PAN validation, identity verification status
- Core Banking System (CBS) — existing account data, existing loan balances
- VRM Service — village risk mapping category
- Blacklist / Debarment Service — internal and RBI defaulter list

9.2 Downstream Systems (Rule Outputs)

- LOS Workflow Engine — route loan application based on BRE outcome (auto-approve / deviate / reject)
- Notification Service — alert credit officers of deviation assignments
- Reporting / MIS — rule execution summary for portfolio monitoring
- Audit Service — immutable execution log for compliance

10. Assumptions & Constraints

10.1 Assumptions

- All upstream data services expose REST APIs with SLA < 200ms
- Redis is available as shared cache infrastructure
- Kafka is available for async rule evaluation events
- PostgreSQL is the primary data store for rule definitions and execution logs

- The BRE will be deployed as a microservice within the existing Spring Boot ecosystem

10.2 Constraints

- Rules must be evaluated within 500ms SLA for synchronous calls
- No third-party rule engine licenses (e.g. Drools is acceptable as open source; IBM ODM is not)
- Rule changes in ACTIVE state must go through approval workflow — direct edits not permitted
- All PII data in rule inputs must be masked in execution logs

11. Acceptance Criteria

- AC-001: A business analyst can create and activate a new rule without any code deployment
- AC-002: Rule evaluation API returns correct PASS/DEVIATION/FAIL for all defined test cases
- AC-003: All rule executions are persisted with full input/output audit trail
- AC-004: Deviation routing correctly escalates to configured role (CU_L3/CU_L4/CREDIT_HEAD)
- AC-005: System handles 1,000 concurrent rule evaluation requests within 500ms P99
- AC-006: Rule version rollback restores previous rule behavior within 60 seconds
- AC-007: Rule simulation correctly predicts production outcome for 100% of test cases

12. Glossary

Term	Definition
BRE	Business Rules Engine — system that evaluates predefined business logic against input data
MFI	Microfinance Institution — lender serving low-income borrowers typically through group lending
SHG	Self Help Group — group of borrowers jointly liable for loan repayment
VRM	Village Risk Mapping — risk classification of a geographic village/area
FOIR	Fixed Obligation to Income Ratio — ratio of fixed EMI obligations to gross income
DSCR	Debt Service Coverage Ratio — net operating income divided by total debt service
LTV	Loan to Value — ratio of loan amount to collateral value for secured loans
PEP	Politically Exposed Person — individual with prominent public function, higher AML risk
Deviation	Rule outcome where loan is eligible with exceptions, routed to a higher authority for manual approval